Ibrahim Ahmad Siddiqi

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SUMMARY

AI application engineer and Software Engineering graduate (FAST-NUCES, 2026). I build LLM features into backends: an LLM tool that turns requirements documents into test cases (Systems Limited), the AI service of a content-automation platform (live web research, RAG over Qdrant, Gemini with Groq fallback, automatic repair of malformed JSON), and the orchestrator for a 13-agent system. Python/FastAPI, Node.js, PostgreSQL.

TECHNICAL SKILLS

- **AI / LLM:** Gemini and Groq (Llama) APIs, RAG pipelines, multi-agent orchestration, Sentence-Transformers embeddings, prompt design, JSON-mode output, output validation and repair
- **Backend:** FastAPI, Express.js, Flask, REST APIs, Prisma, JWT auth, OAuth 2.0, rate limiting, Pydantic
- **Languages:** Python, JavaScript (Node.js), SQL
- **Databases & Tools:** PostgreSQL, MongoDB, SQLite, Qdrant, ChromaDB, Git, Docker, AWS Rekognition, Pytest

EXPERIENCE

Full-Stack Python Intern, Research Lab, FAST-NUCES (Remote) **06/2026 – 08/2026**

- Built **EcoTrack**, a PyQt6 desktop app that samples per-process CPU, GPU, RAM and network usage every 5 s (psutil, NVIDIA NVML) and converts it to CO2 estimates using IEA grid emission factors for 11 regions.
- Built the full pipeline: SQLite time-series storage, live dashboard with history and alerts, iterated on review feedback.

Software Engineering Intern, Systems Limited, Islamabad **06/2025 – 08/2025**

- Built **SPECmate**, reviewed by the Software engineering team: an LLM-powered generator (Llama 3 70B via Groq) that turns raw requirements documents into test cases, use cases and functional modules.
- Engineered a multi-layer validation engine (AI semantic checks, schema integrity, data normalization) on an Express.js, SQLite and React stack, refined through review cycles.

PROJECTS

SEOmation: AI SEO Content Automation Platform *(FastAPI, Gemini, Groq, Qdrant)* **08/2025 – 05/2026**

Final-year project, team of 3. I built the backend and AI service.

- Built the FastAPI AI service: each request runs live web search and scraping, adds RAG context from Qdrant (namespaced per user & niche), then generates platform-specific JSON with Gemini, falling back to Groq on failure.
- Validated every LLM response (JSON structure, focus keyword, meta length, output language) and sent failures through a one-shot repair pass instead of failing the request; scored drafts for grammar and readability.
- Built the Express/Prisma backend around it: JWT auth, rate limiting, OAuth publishing to WordPress, LinkedIn and Instagram with encrypted tokens, and a retrying scheduler persisted in PostgreSQL.

Multi-Agent AI Orchestration System *(FastAPI, Gemini API, ChromaDB)* **09/2025 – 11/2025**

- In a 3-person team, built the central orchestrator for a 13-agent educational system (agents built by other teams): one Gemini call handles intent detection, parameter extraction and clarifying questions, replacing multi-step routing and cutting LLM calls per request; requests go only to agents that pass a health check.
- Added a two-tier memory layer: short-term session recall in the orchestrator and a per-agent long-term store.

FYP Handbook RAG Q&A *(Python, Sentence-Transformers, FAISS, Groq)* **11/2025**

- In a 2-person team, built the RAG pipeline: chunked and embedded the handbook with Sentence-Transformers, retrieved top-K chunks by cosine similarity in FAISS, and generated answers with page citations via Groq.
- Added an embedding cache (repeat queries in ~1 ms vs ~1.2–1.4 s cold), exponential-backoff retries, and global and per-user rate limits.

Image Moderation App *(FastAPI, MongoDB, AWS Rekognition, JWT, Docker)* **05/2025**

- Built a role-based moderation API: admins issue and revoke user tokens (JWT, stored in MongoDB), usage is logged per token, and AWS Rekognition flags unsafe images (correct on 95 of 100 test images); runs via Docker.

EDUCATION

BS Software Engineering, FAST-NUCES, Islamabad **2022 – 06/2026**

Courses: Generative AI, Applied AI, Software Design & Architecture, Databases, Data Structures & Algorithms

CERTIFICATIONS & ACTIVITIES

Certifications: Generative AI for Software Development, AWS Cloud Quest Cloud Practitioner

Activities: Technical Workshop Team, Google Developer Student Club (09/2024 – 04/2025)